Project Development Phase

Date	20 October 2026
Team ID	
Project Name	Global Olympic Sports Performance Analysis using Tableau
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Downloaded the "Olympic Games Performance & Medal History 1992–26" dataset from Kaggle (4 CSV files).
- Connected athletes.csv, medal_results.csv, games_summary.csv and alltime_country_rankings.csv to Tableau Public using Connect → Text file.
- Confirmed row counts: 8,000 / 910 / 18 / 60 (8,988 rows in total).

Step 2: Create Relationships

- alltime_country_rankings → medal_results on *noc_code*.
- alltime_country_rankings → athletes on *noc_code* (athletes has no year column).
- medal_results → games_summary on *year* and *season* (1992 had both Summer and Winter Games).
- Relationships were used instead of joins to avoid double-counting medals.

Step 3: Check for Missing Values and Duplicates

- Checked key fields (country, noc_code, region, year, season, medal counts) for nulls; none were found.
- Verified that every country–Games row in medal_results is unique.

Step 4: Validate Data Types

- Country / Host Country → Geographic role (Country/Region)
- Year → Number, converted to **Discrete Dimension**
- Gold, Silver, Bronze, Total Medals, Athletes Sent → Number (Measure)
- Medal Efficiency Pct → Number (decimal)
- Host Nation (0/1) → converted to Dimension for filtering
- Region, Season, Sport Category, Sports Investment → String

Step 5: Verify Data Consistency

- Totals cross-checked: 6,025 medals in medal_results match games_summary and alltime_country_rankings.
- Field names taken from the correct table for each chart to avoid wrong totals.

Step 6: Prepare Calculated Fields

Field	Formula
Efficiency Ratio	AVG([Medal Efficiency Pct]) / 100
Remaining	1 - [Efficiency Ratio]
Medals per Athlete	SUM([Total Medals]) / SUM([Athletes Sent])
Gold Share %	SUM([Gold]) / SUM([Total Medals])
Host Advantage	IF [Host Nation] = 1 THEN "Host Nation" ELSE "Non-Host" END
Olympic Edition	STR([Year]) + " " + [Season]
Selected Medals	CASE [Select Medal] WHEN "Gold" THEN SUM([Gold]) WHEN "Silver" THEN SUM([Silver]) WHEN "Bronze" THEN SUM([Bronze]) ELSE SUM([Total Medals]) END
Parameter: Select Medal	String list: Total, Gold, Silver, Bronze

Step 7: Final Clean Dataset

- All four tables related, validated and ready for visualization in Tableau Public.

Business Questions with Visualization

S.No.	Business Question	Visualization
1.	How many medals, nations, athletes and Games are covered?	KPI Cards (Text)
2.	Which countries have won the most medals?	Total Medals by Country (Horizontal Bar, Top 15, Select Medal parameter)
3.	Which sport categories give the highest medals per athlete?	Avg Medals per Athlete by Sport Category (Horizontal Bar)
4.	How have total medals changed over the years?	Trend of Total Medals Over the Years (Line Chart, Summer vs Winter)
5.	Where are medals concentrated geographically?	Total Medal Achievements Across Nations (Filled Map)
6.	What share of medals does each region win?	Medal Share by Region (Donut Chart)
7.	How are medals split across regions and countries?	Medals by Region and Country (Treemap)
8.	Do host nations perform better at home?	Medal Totals Across Olympic Hosts (Horizontal Bar)
9.	Is sports investment linked to medal success?	Sports Investment vs Medal Performance (Packed Bubbles)
10.	Which countries are stronger in Summer or Winter?	Summer vs Winter Medals by Country (Connected Dot Plot)
11.	What is the overall average medal efficiency?	Average Medal Efficiency (Radial Gauge)
12.	Which regions are most efficient?	Avg Medal Efficiency by Region (Pie Chart)

Visualizations

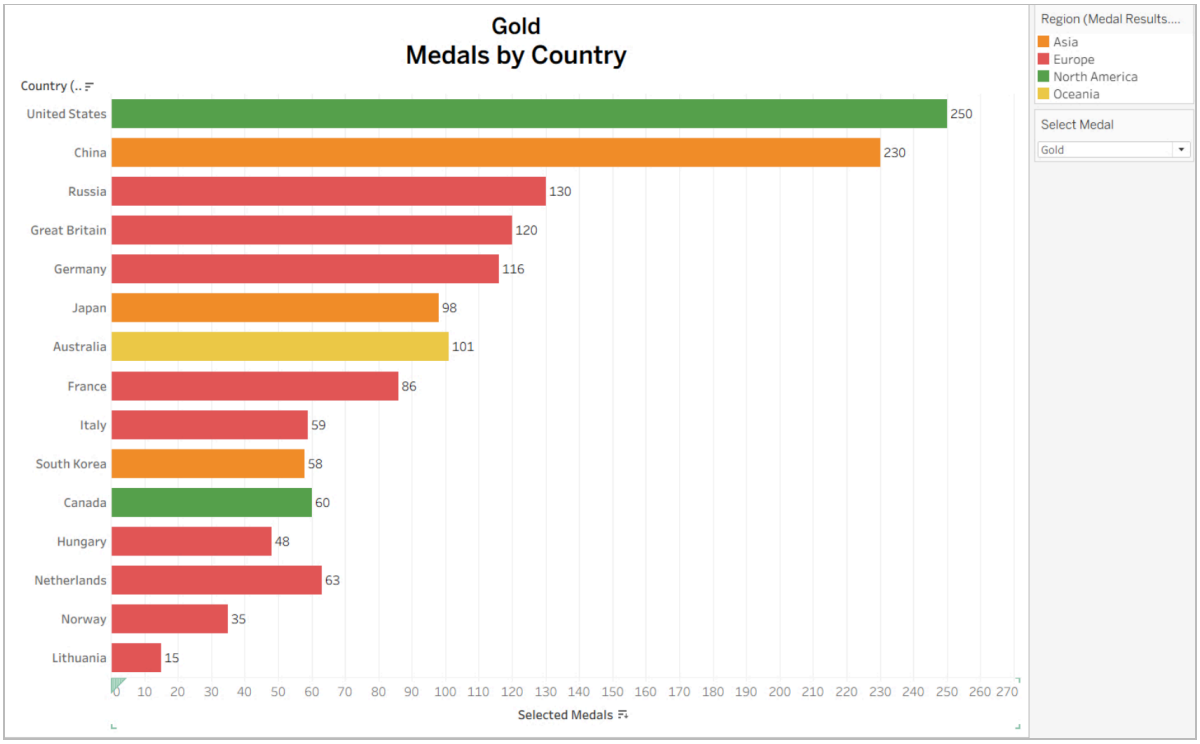


Figure 1: Total Medals by Country (Horizontal Bar) – Select Medal parameter set to Gold

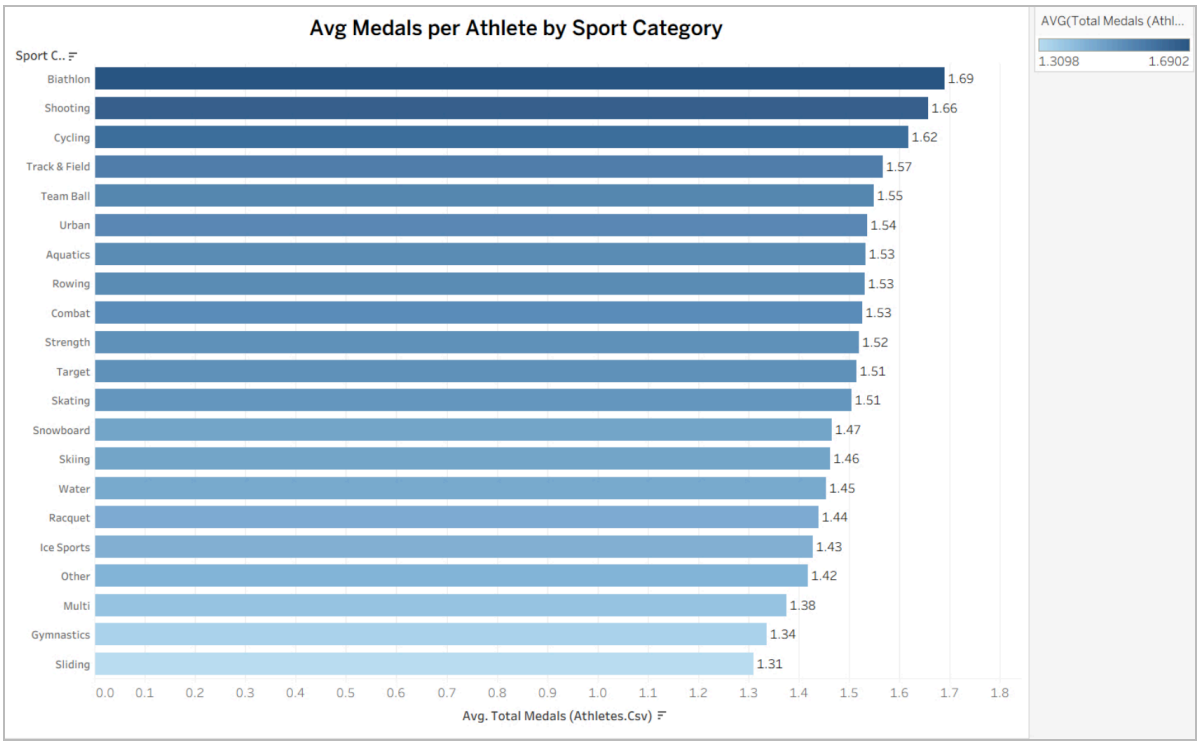


Figure 2: Avg Medals per Athlete by Sport Category (Horizontal Bar)

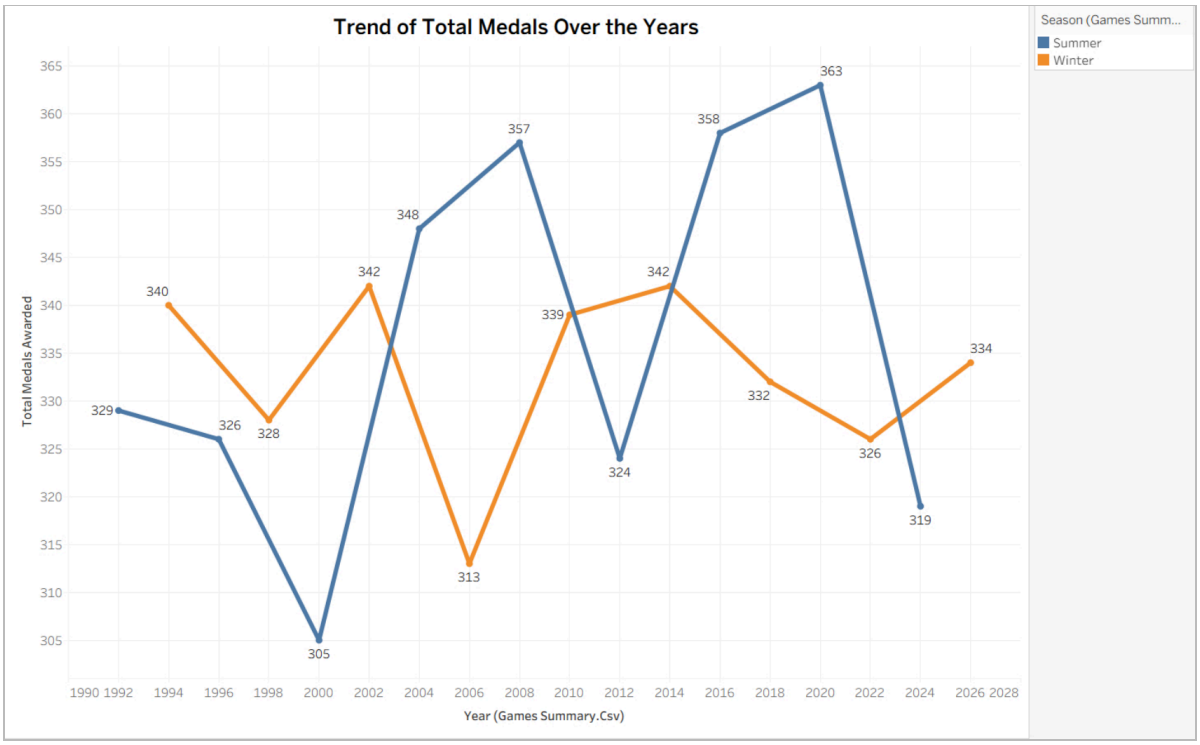


Figure 3: Trend of Total Medals Over the Years (Line Chart)

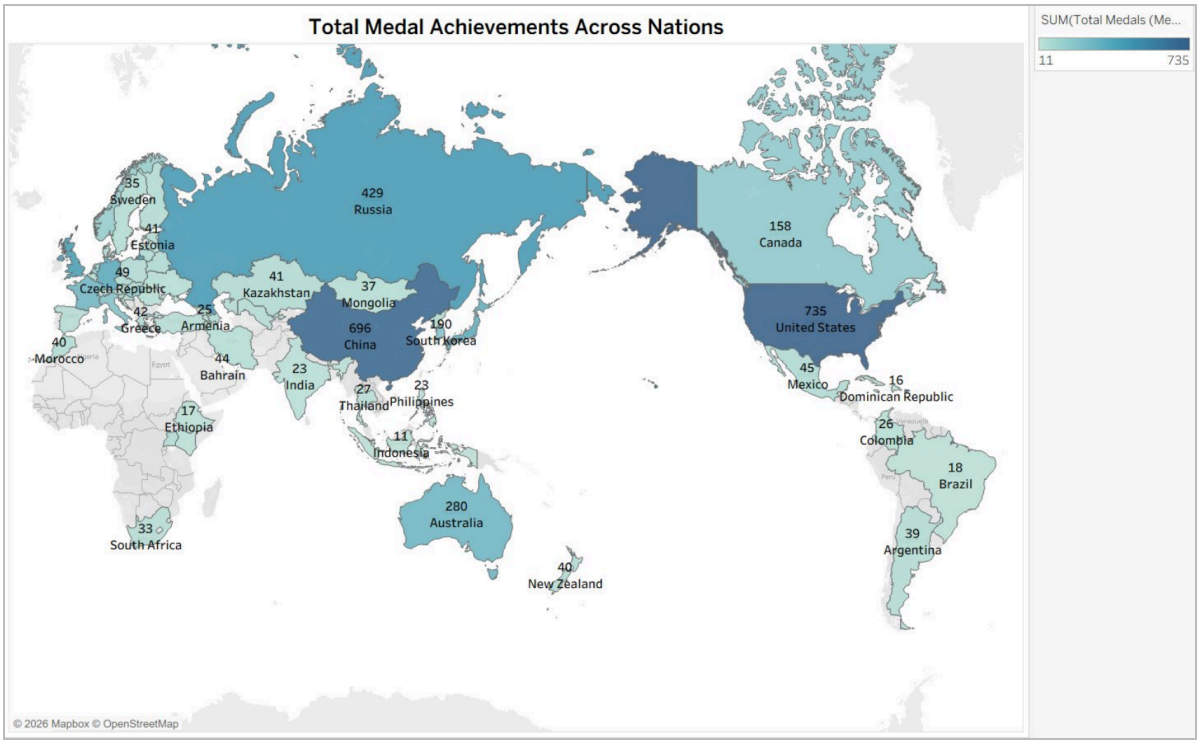


Figure 4: Total Medal Achievements Across Nations (Filled Map)

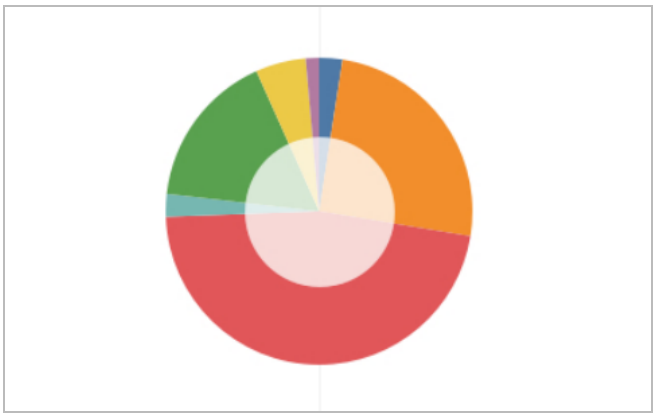


Figure 5: Medal Share by Region (Donut Chart)

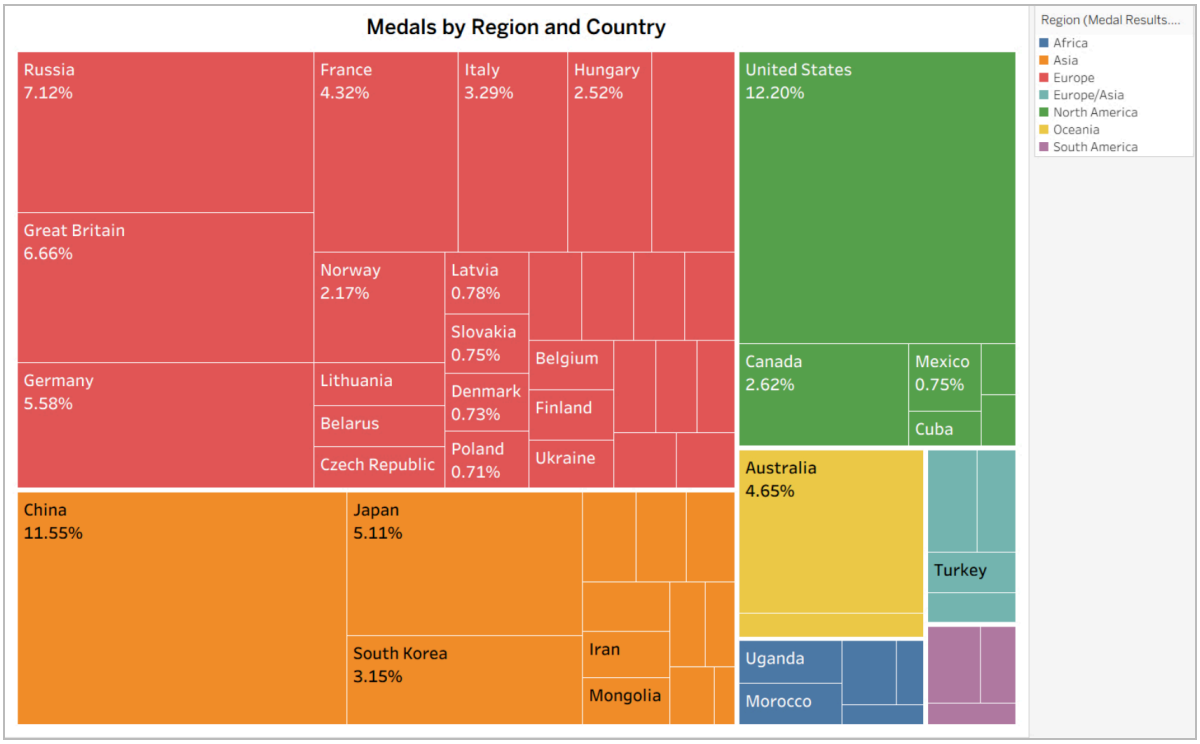


Figure 6: Medals by Region and Country (Treemap)

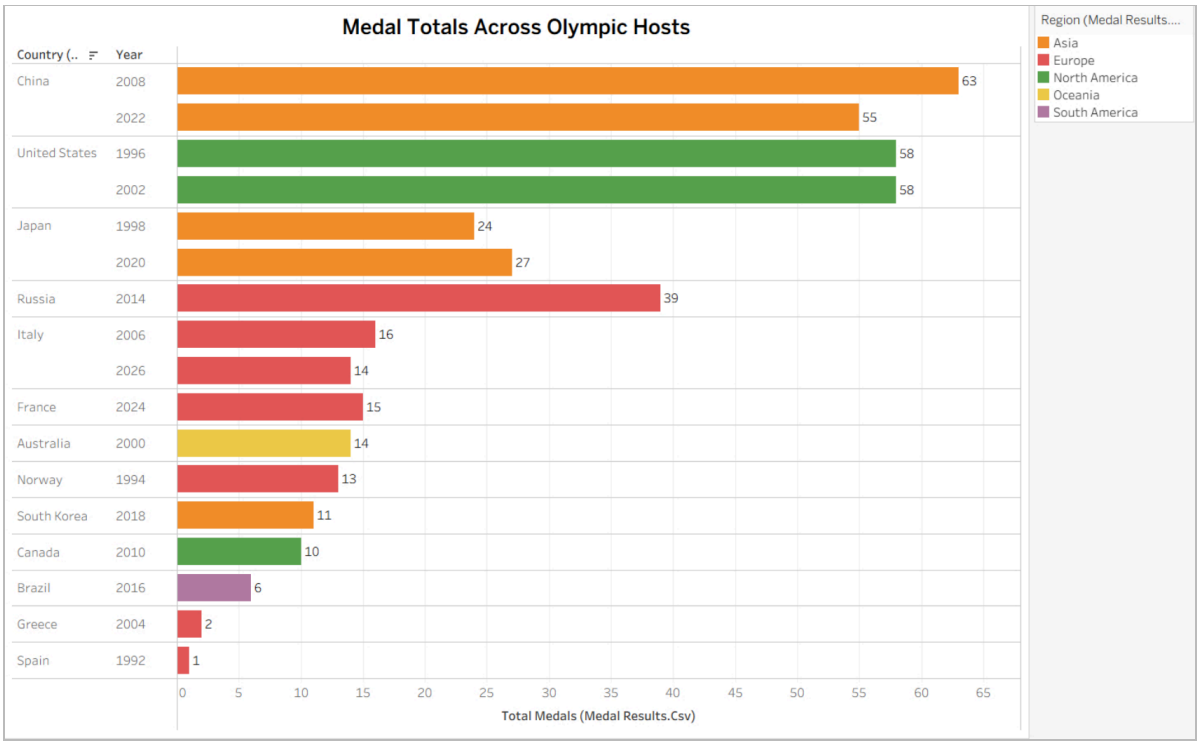


Figure 7: Medal Totals Across Olympic Hosts (Horizontal Bar)

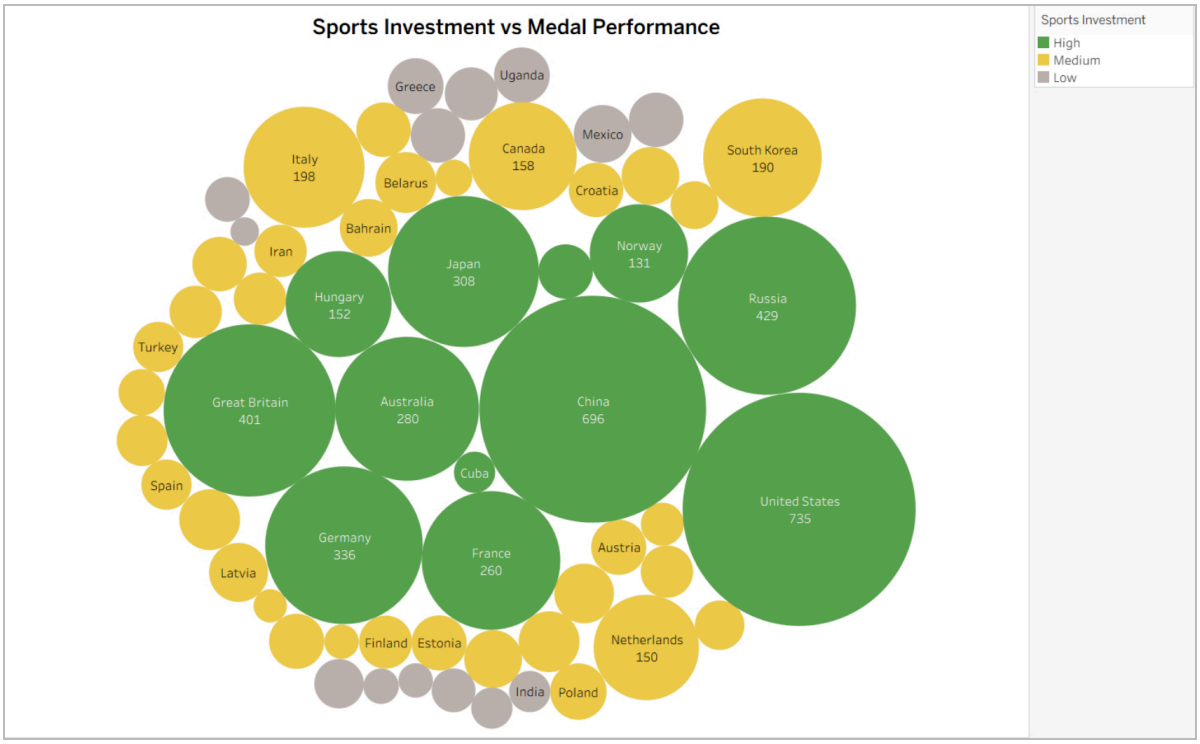


Figure 8: Sports Investment vs Medal Performance (Packed Bubbles)

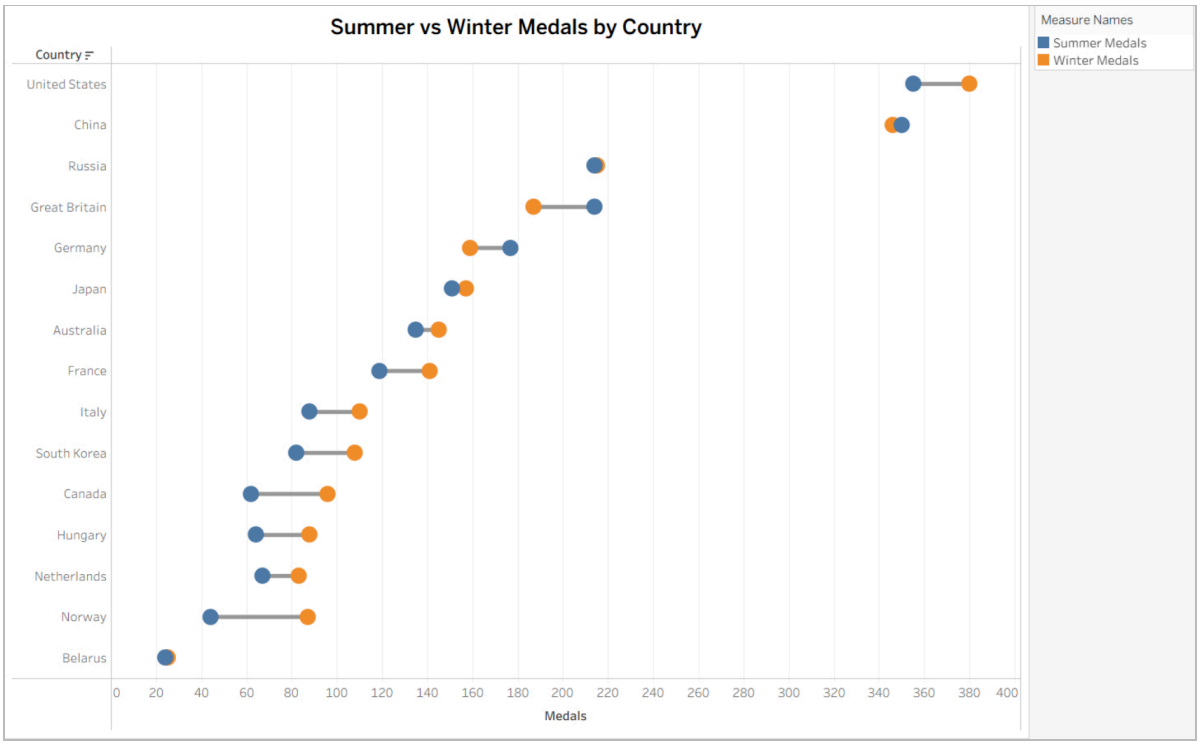


Figure 9: Summer vs Winter Medals by Country (Connected Dot Plot)

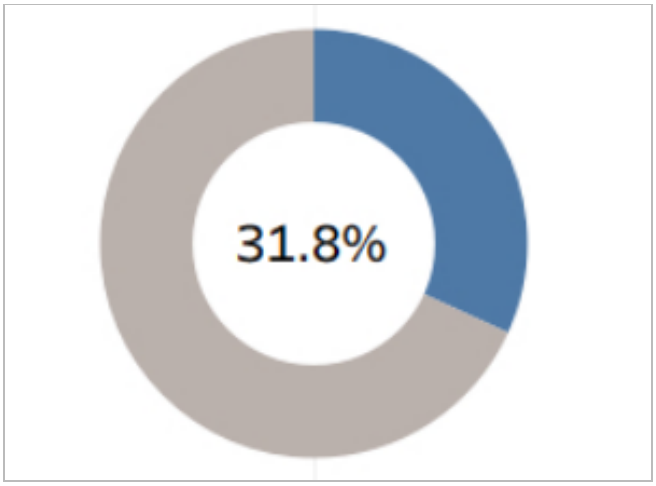


Figure 10: Average Medal Efficiency (Radial Gauge)

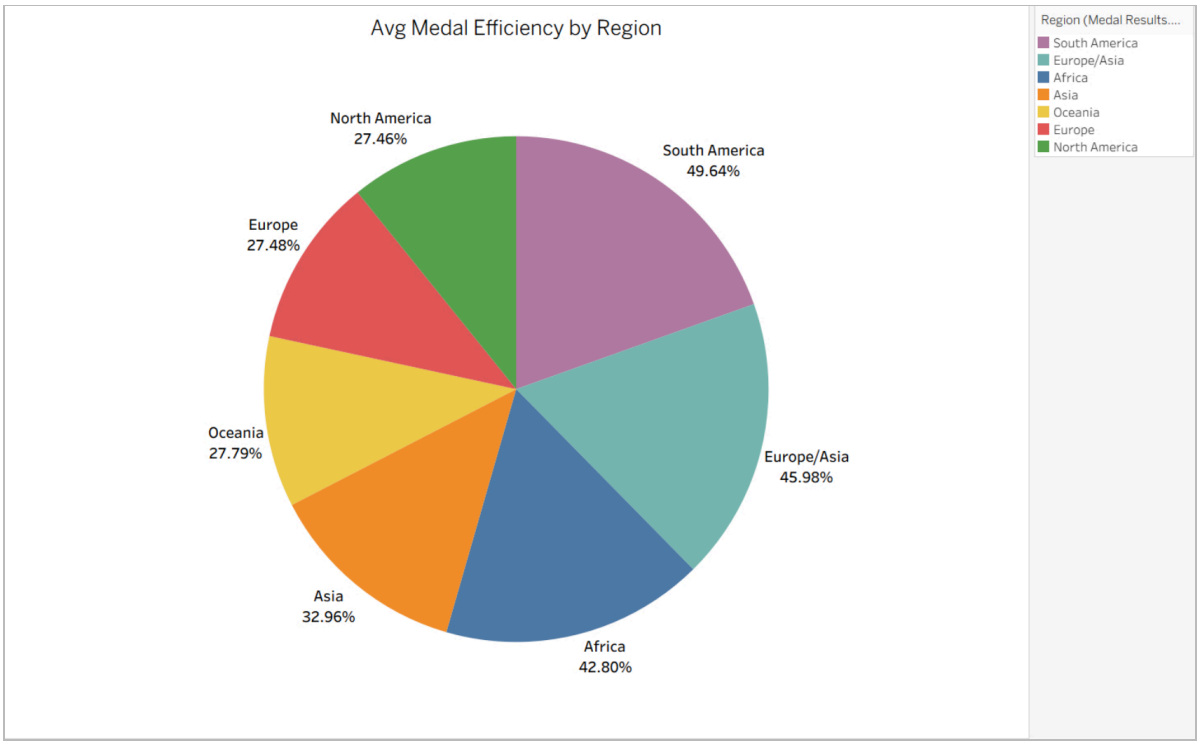


Figure 11: Avg Medal Efficiency by Region (Pie Chart)



Figure 12: KPI Cards – Total Medals, Nations, Athletes, Olympic Games